

Evaluated System Inputs and Ground Truth Datasets
18 Semantic Workload Families · 83 Prompt Variants · Administrator-
Controlled Catalog (4 Digest-Pinned Images) · Static K8s Profiles

**Stream E1 & E2:
Offline Quality
& Robustness**

Macro-level analysis over 18
semantic workload families

**Offline Bench-
mark Harness**

Dual-system runner (P1
vs P2) · Seeded execution
· Bootstrap sampling

Descriptive Metrics

JointAccept@1 · Pro-
file/Image Acc. · Constraint
Violations · Macro SRR

Stream E1 / E2: Offline Quality

**Stream E4: Local
Cluster Resource Trials**

Kubernetes cluster trials
on OrbStack environment

Kubernetes Pod Runner

421 real Pod trials · Dynamic
cgroups v2 monitoring
(100ms) · OOM detection

Empirical Observations

98.1% success (413/421)
· 1.9% OOM (8) · Safe
dynamic envelope vs static

Stream E4: Cluster Resources

**Stream E5: Capabil-
ity & OCI Storage**

Container sandbox verification
& OCI layer accounting

Dual Verification Engine

Isolated Docker probe execution
(166 checks) · Manifest
layer digest hash analysis

**Capability &
Layer Results**

100% probe pass (eligible)
· 11.39 GB logical vs 8.25
GB unique (27.61% reuse)

Stream E5: Image & Storage

**Stream E3: Hu-
man Study Protocol**

User interaction
evaluation protocol

Harness & Protocol

Within-subject counter-
balanced design (E3)

Evidence Status

E3: Protocol only
(no participants)

Stream E3: User Study Protocol